

Rust (in production) Without Fear

RustConf 2026

Victor Ciura

Principal Rustacean

Rust Tooling @ [Microsoft](#)

Early incubations → Rust projects in prod

Microsoft timeline: From early incubations to premier Rust projects in production ([today](#))

- Project Mu
- Pluton security processor
- SymCrypt + rustls
- Azure Integrated HSM
- Azure Boost Agents
- Open VMM / Open HCL
- Hyper-V
- Azure Data Explorer
- Drasi
- MIMIR
- Caliptra - Hardware Root of Trust
- Hyperlight / WASM
- Azure SDK for Rust
- Oxidizer SDK
- Rust @ Edge / Chromium
- DirectWrite, GDI
- Windows-rs
- Windows-drivers-rs
- Azure services
- M365 services
- Copilot stack
- ...



Copilot SDK + many AI-related core tools & services (powered by Rust)

The Journey to "Tier-1 language"

- Today: Rust sits among C++, C#, TypeScript as the best supported languages for internal development at Microsoft
- What does it take to make Rust prod-ready at MS?
- Compliance and SDL requirements (Windows + Linux)
- Secure Supply Chain builds
- A long list of ES tools & workflows that needed to be made Rust-aware
- Quality workflows (test, cov, san, SA, CodeQL)
- Developer tooling
- Great Windows ecosystem compatibility

Paved Path for Rust



1ES: Paved Path for Rust engineering systems



Crate security in enterprise environment

- Internal crate registry (feeds, policies, quarantine, docs)
- Crate publishing workflows (public, private)
- Crate review/evaluation & audit



Microsoft Rust SDK

- Toolchain acquisition (Dev / CI)
- Compiler toolchains (various hosts/targets), ancillary dev tools
- Support hybrid projects (C++/Rust/C#)
- ES workflows & services
- Release qualification testing
- Docs, samples, guidance, skills
- Servicing & longer support cycles



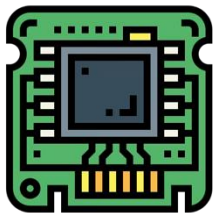
Agentic workflows Rust-aware

Introducing “Rust/C2”

- The need for a [Windows-first compiler codegen](#) for Rust
- An alternative rustc backend for Windows
 - `codegen_llvm`, `codegen_gcc`, `codegen_craneflight`, ... `codegen_utc`
 - `rustc` + `codegen_utc` + `C2` (`MSVC` backend)
- [Native Platform Evolution](#) (unified codegen platform for C++ and Rust)
- Seamless [Rust/C++ interop](#) for hybrid projects
- [Unified](#) code-cov and [SPGO](#)
- [Cross-language](#) inlining and optimization
- Post-link [compliance](#) and [servicing](#) (APIscan, Hotpatch)
- [Binary hardening](#) and [code security](#) features (SDL)
- Enhanced [DevEx](#) and [Debugging](#)
- [Production-ready](#) since earlier this year
 - [Self hosted](#) since Rust 1.90 on x64/aarch64 [Windows](#)
 - Already [over 100 MS project repos](#) build with our new codegen backend (more switching every week)

Microsoft Cloud runs on Rust

- Azure infra & core services
 - [Azure SDK](#) for Rust
- M365 services
 - [Oxidizer SDK](#)
- [Copilot](#) tech stack & services
- Bare-metal, firmware, drivers, Linux, Windows



2026 - Are we Rust-ready yet?

- YES, from Edge to Cloud, Rust is everywhere at Microsoft!
- What's **your** Rust journey?
- A bunch of **MS Rustaceans** are here on-site 🙌
- Let's talk!